

Characteristics of Nucleus

- ❖ The nucleus of an atom is its center. Most of the mass of an atom is concentrated in the nucleus.
- ❖ One atomic mass unit is defined as $(1/12)^{\text{th}}$ of the mass of the $^{12}_6\text{C}$ isotope. It is represented by the symbol u and it is the average mass of a nucleon.
- ❖ The stability of any nucleus depends on the number of protons and neutrons. For small nuclei to be stable, the number of protons must be roughly equal to the number of neutrons. As the number of protons increases, however, more neutrons are needed to maintain stability.
- ❖ Nuclei of different elements have different sizes because the mass number (A) for different elements is different. Density of nucleus is the almost same for all elements.
- ❖ Average radius of nucleus may be written as,

$$R = R_0 A^{1/3}, \quad R_0 = 1.1 \times 10^{-15} \text{ m}$$

Mass Defect

The sum of the individual masses of the separated protons and neutrons exceeds the mass of the stable nucleus by an amount ΔM . This difference in mass is known as the **mass defect** of the nucleus.

Mass-energy Relation

According to mass-energy relation, the mass of a body is a measure of its energy content.

Equivalence of mass and energy, $E = mc^2$

Note: $1 u = 1.66 \times 10^{-27} \text{ kg} \equiv 931.5 \text{ MeV}$

Binding Energy

- ❖ The binding energy of a nucleus is the energy that would have to be provided to split a nucleus into its individual nucleons. Binding energy of nucleus ${}_Z^AX^A$ of mass M , is given by $BE = (Zm_p + Nm_n - M)c^2$ and the mass defect, $\Delta m = Zm_p + Nm_n - M$.
- ❖ The average energy required to remove a nucleon from the nucleus to infinite distance is known as the **binding energy per nucleon**.
- ❖ Nuclei with high binding energies per nucleon are very stable as it takes a lot of energy to split them. Nuclei with lower binding energies per nucleon are easier to split.

- ❖ In order to become more stable, unstable nuclei tend to release some of their energy. Releasing this energy would decrease the amount of energy they contained, and therefore increase the amount of energy that must be added to them to split them apart.

Q Value

- ❖ Energy released in any nuclear reaction or collision is called Q value of the reaction.
- ❖ For nuclear reaction, $A + B \rightarrow C + Q$ (Energy)
The energy of reaction Q is given by,

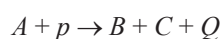
$$Q = (m_A + m_B - m_C)c^2 = BE_C - BE_A - BE_B$$
- ❖ If Q is positive, energy is released and products are more stable in comparison to reactants.
If Q is negative, energy is absorbed and products are less stable in comparison to reactants.

Nuclear Force

- ❖ The strong nuclear force is the force that holds nucleons together in the nucleus of an atom. It acts only over very short distances.
- ❖ Inside a nucleus, the nucleons are very close. The pull of the strong nuclear force is much greater than the push of the protons repelling each other, and therefore the nucleus remains intact.
- ❖ Adjacent neutrons experience no electrostatic repulsion between each other. There is only the strong force of attraction between them.
- ❖ Two protons do repel each other when they are brought together, but in the nucleus they are so close to each other that the force of repulsion is overcome by the even stronger nuclear force.

Nuclear Fission

- ❖ By bombarding a particle on a heavy nucleus ($A > 230$), it splits into two or more light nuclei. In this process certain mass disappears which is obtained in the form of energy (enormous amount)



- ❖ The energy released in a fission reaction comes from the difference between the mass of the original nucleus and the combined mass of the fission fragments.
- ❖ A chain reaction occurs when neutron, emitted from the decay of one atom, initiate fission in surrounding nuclei.
- ❖ An uncontrolled chain reaction occurs when every free neutron goes on to produce another fission reaction. It occurs in nuclear bombs and releases an enormous amount of energy.
- ❖ A controlled chain reaction occurs when only some of the emitted neutrons are able to induce further fission reactions, and the remaining neutrons are absorbed by a material that is not fissionable. In this situation energy can be released at a constant rate.
- ❖ Enrichment is the process of increasing the percentage of ^{235}U in a sample of uranium. Enrichment is important because naturally occurring uranium does not have a high enough percentage of ^{235}U to sustain a chain reaction.

Nuclear Fusion

- ❖ It is the phenomenon of fusing two or more light nuclei to form a single heavy nucleus.
 $A + B \rightarrow C + Q$ (Energy)
- ❖ The product (C) is more stable than reactants (A and B) and $m_C < (m_A + m_B)$.
- ❖ Mass defect, $\Delta m = [(m_A + m_B) - m_C]$ amu
- ❖ Energy released is $E = (\Delta m) \times 931 \text{ MeV}$

Radioactivity

A nuclear phenomenon in which an unstable nucleus undergoes a decay is known as radioactivity. A decay equation is a representation of a decay reaction. It shows the changes occurring in nuclei undergoing decay and lists the products of the decay. The nucleus remaining after an atom undergoes radioactive decay is called a daughter nucleus. The daughter nucleus is more stable than the original nucleus. Generally, there are three types of radioactive decays,

- α decay
- β^- and β^+ decay
- γ decay

α decay

- ❖ An α -particle is a relatively slow-moving decay product consisting of two protons and two neutrons. It is the nucleus of helium and so can be written as ${}^4_2\text{He}$. α -particles carry positive charge.
- ❖ α -decay process: ${}^A_Z\text{X} \longrightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\text{He}$
- ❖ Q -value is, $Q = [m({}^A_Z\text{X}) - m({}^{A-4}_{Z-2}\text{Y}) - m({}^4_2\text{He})]c^2$
- ❖ For the above shown decay kinetic energy of α -particle will be given by, $K_\alpha = \frac{m_Y}{m_X}Q$

β^- and β^+ decay

- ❖ A β -particle is a fast-moving electron that is ejected from an unstable nucleus. The electron is produced when a neutron transforms into a proton and an electron.

- ❖ In β -decay the resulting daughter nucleus has the same number of nucleons as the parent, but has one less neutron and one more proton.
- ❖ β -particles are very light when compared to α particles. They travel at a large range of speeds—from that of an α -particle up to 99% of the speed of light.

β^- decay



$$Q\text{-value} = [m({}^A_Z\text{X}) - m({}^A_{Z+1}\text{Y})]c^2$$

β^+ decay



$$Q\text{-value} = [m({}^A_Z\text{X}) - m({}^A_{Z-1}\text{Y}) - m_e]c^2$$

where $m({}^A_Z\text{X})$, $m({}^A_{Z-1}\text{Y})$ are atomic masses

- ❖ Electron capture (K-capture): When atomic electron is captured by nucleus, X-rays are emitted.



$$Q\text{-value} = [m({}^A_Z\text{X}) - m({}^A_{Z-1}\text{Y})]c^2$$

where $m({}^A_Z\text{X}) - m({}^A_{Z-1}\text{Y})$ are atomic masses.

γ decay

A γ -ray is the packet of electromagnetic energy released when a nucleus is formed in excited state after α or β -decay which releases energy to come down to ground state. γ -rays travel at the speed of light, carry no mass or charge, and are not deflected by electric or magnetic fields.

Law of Radioactive Decay

The rate of disintegration is directly proportional to the number of radioactive atoms present at that time i.e., rate of decay $\propto$ number of nuclei.

$$\text{Rate of decay} = \lambda (\text{number of active nuclei})$$

$$\text{i.e., } \frac{dN}{dt} = -\lambda N.$$

where λ is called the decay constant.

If N_0 is the number of parent nuclei at $t = 0$. The number that survives at time t is $N = N_0 e^{-\lambda t}$

$$\text{Probability of a nucleus for survival of time } t = \frac{N}{N_0} = \frac{N_0 e^{-\lambda t}}{N_0} = e^{-\lambda t}$$

Half-life

- ❖ A **half-life** is the time taken for half of a group of unstable nuclei to decay. In other words, it is the time during which the number of undecayed atoms in the sample becomes half the total number of atoms present initially in the sample. Half-life is represented by the symbol $T_{1/2}$.

$$\text{Half life: } T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

- ❖ Number of nuclei present after n half lives i.e. after a time $t = nT_{1/2}$, $N = \frac{N_0}{2^n}$

- ❖ A radioactive nucleus can decay by two different processes having half lives t_1 and t_2 respectively. Effective half-life of nucleus will be given by $\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$.

Mean or Average Life

The ratio of the total life time of all the atoms of the element to the total number of atoms present initially in the sample of the element is known as the **mean or average life**. Mean life is represented by the symbol τ . It is equal to the reciprocal of the decay constant (λ) of the element.

$$\text{Average life: } \tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693}$$

Activity

- ❖ Activity of sample:
 $R = \lambda N = R_0 e^{-\lambda t}$
- ❖ Unit: 1 Bq = 1 decay/s, 1 curie = 3.7×10^{10} Bq, 1 rutherford = 10^6 Bq
- ❖ Activity per unit mass is called specific activity.
- ❖ Activity after n half lives: $\frac{A_0}{2^n}$